



United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Final Exam Summer 2024

CSE 4611/CSI 411: Compiler Design/Compiler

Total Marks: 50 Duration: 2 hours

Answer all questions. Figures in the right-hand margin indicates full marks.

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

1. Read the following code and detect different types of errors.

[10]

```
1. #include <stdio.h>
2. void checkEvenOdd(int num)
3. void displayMenu(int option)
4. int mian() {
5.     int option,num;
6.     print("Enter a number: ");
7.     scanf("%d", number);
8.     if (num < 0) {
9.         printf("The number is positive.\n");
10.    } else if (num > 0) {
11.        printf("The number is negative.\n");
12.    } else {
13.        printf("The number is zero.\n");
14.    }
15.    checkevenOdd(num);
16.    printf("Choose an option (1-3):\n");
17.    scanf("%f", &option);
18.    switch(option) {
19.        case 1:
20.        case 2:
21.            display(option)
22.            brak;
23.        case 3:
24.            printf("Exiting...\n");
25.            break;
26.        default:
27.            printf("Invalid option.\n")
28.    }
29.    return 'c';
30. }
31. void checkEvenOdd(char num) {
32.     if (num % 2 = 0) {
33.         printf("%d is even.\n", num);
34.     } else
35.         printf("%d is odd.\n", num);
36.     }}}
37. }
38. void displayMenu(int option) {
39.     swich(option) {
40.         case 1:
41.             printf("Hello!\n");
42.             break;
43.         cas 2;
44.             printf("World!\n");
45.             break;
46.     }
47. }
```

2. Construct a Non-recursive Predictive Parsing table for the following grammar & also check whether string “2222223145” is accepted or not. The sequence of Moves by Non-recursive Predictive Parser in a table (should have column: stack,input, action). [5]

$E \rightarrow A$
 $A \rightarrow BCD \mid BC$
 $C \rightarrow 3 \mid \epsilon$
 $D \rightarrow 1 \mid 4 \mid 5$
 $B \rightarrow B2 \mid \epsilon$

3. c) $P \rightarrow bPrQ \mid bRRt \mid RrQ \mid ab$
 $Q \rightarrow 3QRi \mid PQ25 \mid xRgPl \mid \epsilon$
 $R \rightarrow 1RQk \mid 3tQPo \mid RRYi \mid \epsilon$

3. For the following grammars, write down the FIRST and FOLLOW sets for each non-terminal. Consider all the terminals are of single character/number/symbol. **Please answer 3(a,b,c (right above)).** [5*3]

a) $A \rightarrow iBa \mid pBDiu \mid qBrC \mid BCtu \mid \epsilon$
 $B \rightarrow B1Ai \mid 2CDu \mid 1ApBg \mid Cx2 \mid \epsilon$
 $C \rightarrow CCrt \mid zDBl \mid BbCq2 \mid nABcDkB \mid \epsilon$
 $D \rightarrow 0D1B \mid C2x \mid DDB2i \mid 2B2ACi \mid \epsilon$

b) $S \rightarrow NaBGla \mid BbGyl \mid 2NqNgS \mid \epsilon$
 $B \rightarrow 2GG3 \mid BxBiSr \mid tSB \mid GkNN \mid \epsilon$
 $N \rightarrow bBgNo \mid rBgB \mid olBaB \mid qGn \mid \epsilon$
 $G \rightarrow 2GS3 \mid GkBBkl \mid 3BSi \mid 4BiB \mid \epsilon$

4. Left Factor the following. [5*2]

(a) $P \rightarrow [2 + 3]i \mid [2 - 2]y \mid [2 + 33]45 \mid 43iop \mid 43kip \mid 43iol \mid 43jle \mid 432ir \mid 422pl \mid 432p4 \mid 422yu \mid 435uit \mid 435uk \mid 43qio \mid 43qlp \mid 43ikol \mid 43kpo \mid 43ki \mid 43k \mid [2yp] \mid 36 \mid 36hy \mid 4$
 (b) $R \rightarrow Artic \mid Arcani \mid Arcame \mid Arcanio \mid Arc \text{ Love} \mid Arcaml \mid Artnet \mid Arcock \mid Arces \mid Arc \text{ is curve} \mid Arcocips \mid Arcockeak \mid Burnish \mid Butter \mid Bur \mid Buter \mid Buteeo \mid Burri \mid Burnit \mid Burnnop \mid But \mid Burrk \mid Buteos \mid bute \mid butto \mid buttop \mid buttkr$

5. Eliminate immediate left recursion from the following grammar. [5*2]

(a) $S \rightarrow Sa \mid aA \mid aB \mid \epsilon$
 $A \rightarrow bb \mid AB \mid AaC \mid AA$
 $B \rightarrow Ac \mid BA \mid BcC \mid bx$
 $C \rightarrow Cd \mid CCdA \mid cCaC \mid x$

(b) $X \rightarrow YZ \mid XyZN \mid PX/YX \mid X/ZY \mid NMc$
 $Y \rightarrow Y*bb \mid YX+Mc \mid Y/Md \mid Y-Ze \mid p*Y*M \mid x \mid YZYM$
 $Z \rightarrow ZZZ+Y \mid Y+Zb \mid !Z*X/M \mid (Z) \mid /ZM \mid Z+Mc$
 $M \rightarrow M(M) \mid aM+(X*Y) \mid Mb \mid N*YZ \mid Z+ZgZ \mid M*bZ \mid M+Mc$
 $N \rightarrow NN-Ze \mid NY*N \mid Y \mid aXY \mid NZYM \mid aNbc \mid NqrNt \mid \epsilon$